2. Convex optimization

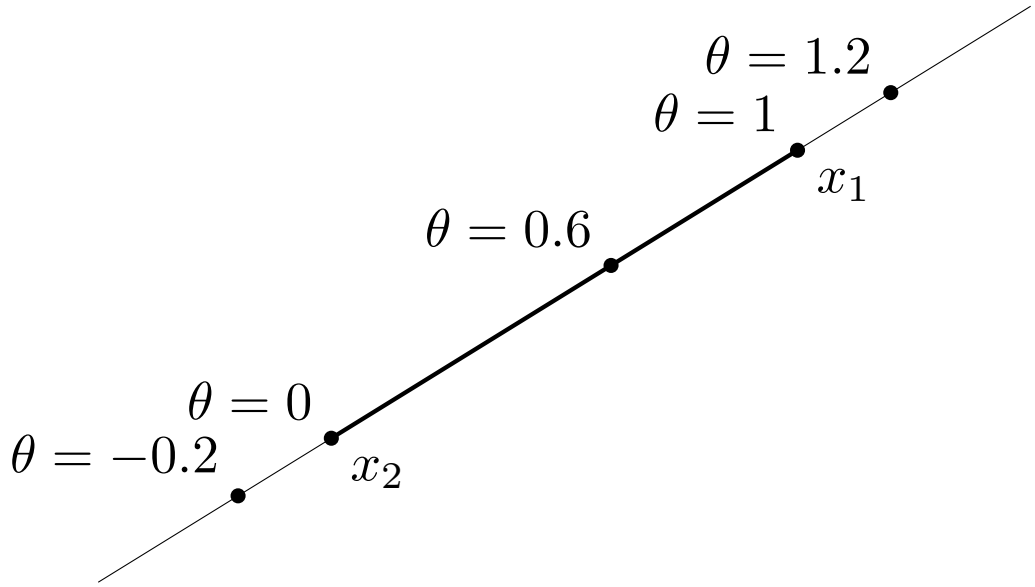
- lines, line segments, and affine sets
- convex sets and cones
- examples and algebra of convex sets
- convex functions and basic properties
- examples of convex functions
- functional operations that preserve convexity
- optimization problems
- equivalent problems and relaxations
- convex optimization problems and optimality conditions
- typical convex optimization problems

Lines and line segments

line through $x_1, x_2 \in \mathbf{R}^n$: all the points

$$y = \theta x_1 + (1 - \theta) x_2 \qquad (\theta \in \mathbf{R})$$

if $\theta \in [0, 1]$, then y is a **line segment** between x_1 and x_2



Affine sets

affine set contains the line through any two points in the set, $i.e.$,

$$x_1, x_2 \in C, \quad \theta \in \mathbf{R} \quad \Longrightarrow \quad \theta x_1 + (1-\theta)x_2 \in C$$

affine combination of points $x_1, \ldots, x_k \in \mathbf{R}^n$:

$$\theta_1 x_1 + \cdots + \theta_k x_k$$

where $\theta \in \mathbf{R}^k$ and

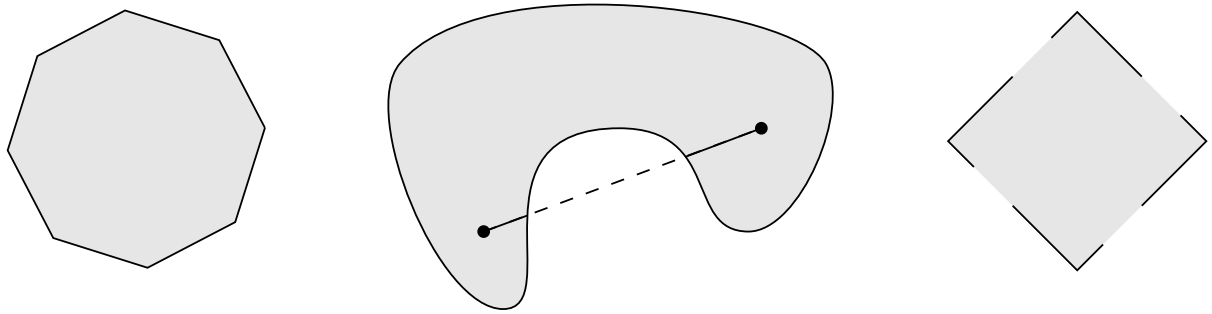
$$\mathbf{1}^T\theta = \theta_1 + \cdots + \theta_k = 1$$

- affine set contains all affine combinations of its points

Convex sets

convex set contains line segment between any two points in the set, $i.e.$,

$$x_1, x_2 \in C, \quad \theta \in [0,1] \quad \Longrightarrow \quad \theta x_1 + (1-\theta)x_2 \in C$$



convex combination of points $x_1, \ldots, x_k \in \mathbf{R}^n$:

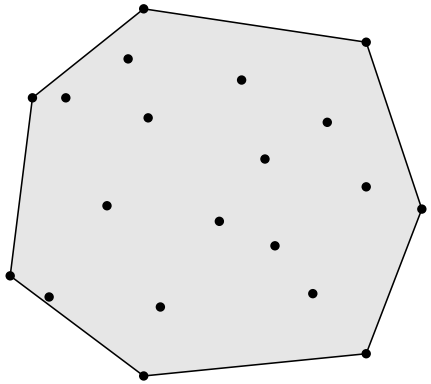
$$\theta_1 x_1 + \cdots + \theta_k x_k$$

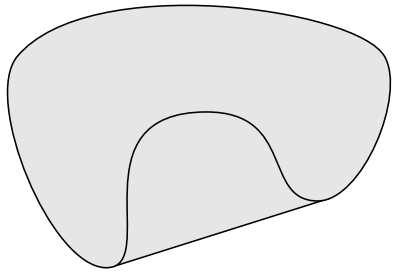
where $\theta \in \mathbf{R}^k$ and

$$\mathbf{1}^T\theta = 1, \quad \theta \succeq 0 \ (i.e., \ \theta_i \geq 0, \ i = 1, \ldots, k)$$

convex hull of set C: set of all convex combinations of points in C, $i.e.$,

$$\mathbf{conv}\, C = \left\{ \sum_{i=1}^{k} \theta_i x_i \;\middle|\; \begin{array}{ll} k \in \mathbf{Z}_{++} & \\ x_i \in C, & i = 1, \ldots, k \\ \mathbf{1}^T \theta = 1, & \theta \succeq 0 \end{array} \right\}$$





Convex combinations of infinite points

let $C \subseteq \mathbf{R}^n$ be a convex set:

- $x_1, x_2, \ldots \in C$, $\theta_1, \theta_2, \ldots \geq 0$, $\sum_{i=1}^{\infty} \theta_i = 1 \implies$

$$\sum_{i=1}^{\infty} \theta_i x_i \in C$$

- $p\colon \mathbf{R}^n \to \mathbf{R}$, $\int_C p(x)\, dx = 1$, $p(x) \geq 0$ for all $x \in C \implies$

$$\int_C p(x) x \, dx \in C$$

- random vector $x \in \mathbf{R}^n$ with $\mathbf{prob}(x \in C) = 1 \implies$

$$\mathbf{E}\, x \in C$$

Convex cones

cone contains all rays through the origin and any point in the set, $i.e.$,

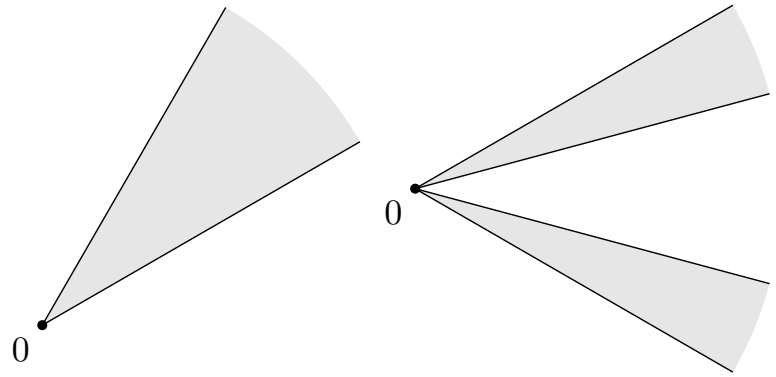
$$x \in C, \quad \theta \geq 0 \quad \Longrightarrow \quad \theta x \in C$$

conic (nonnegative) combination of points $x_1, \ldots, x_k \in \mathbf{R}^n$:

$$\theta_1 x_1 + \cdots + \theta_k x_k$$

where $\theta \in \mathbf{R}^k$ and $\theta \succeq 0$

convex cone contains all conic combinations of its points



Basic examples

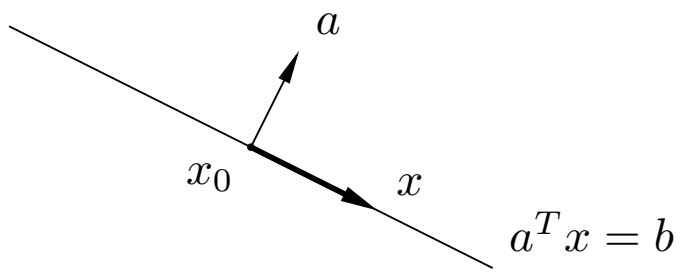
- subspace of $\mathbf{R}^n$ is an affine set and a convex cone, and hence convex
- empty set $\emptyset$, singleton $\{x_0\}$, and $\mathbf{R}^n$ are affine sets
- line through two different points is affine
- line segment between two different points is convex but not affine
- ray $\{x_0 + \theta v \mid \theta \geq 0\}$ $(v \neq 0)$ is convex
  (is a convex cone if contains the origin)
- translation $x_0 + C$ of a convex cone C is convex but not a cone
- solution set of linear equations

$$C = \{x \in \mathbf{R}^n \mid Ax = b\}$$

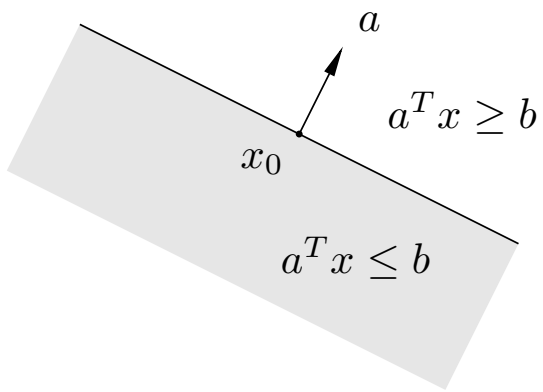
  for $A \in \mathbf{R}^{m \times n}$, $b \in \mathbf{R}^m$ is an affine set

Hyperplanes and halfspaces

hyperplane: set of the form $\{x \in \mathbf{R}^n \mid a^T x = b\}$ $(a \neq 0)$



halfspace: set of the form $\{x \in \mathbf{R}^n \mid a^T x \leq b\}$ $(a \neq 0)$



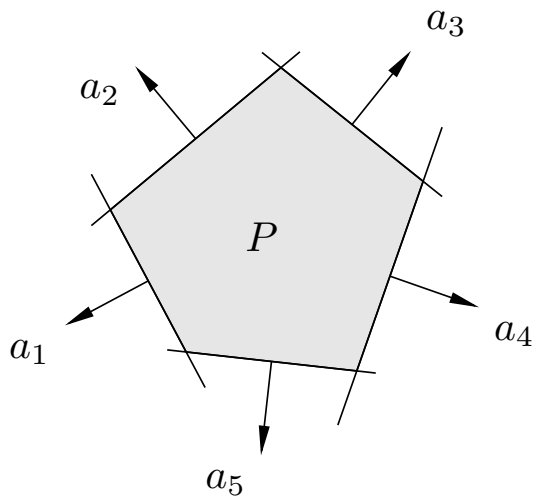
- a is the *normal vector*
- hyperplanes are affine; halfspaces are convex

Polyhedra and simplexes

polyhedron: intersection of finite hyperplanes and halfspaces, $i.e.$,

$$\begin{aligned} P &= \left\{ x \in \mathbf{R}^n \;\middle|\; \begin{array}{ll} a_i^T x \leq b_i, & i = 1, \ldots, m \\ c_i^T x = d_i, & i = 1, \ldots, p \end{array} \right\} \\ &= \{ x \in \mathbf{R}^n \mid Ax \preceq b, \quad Cx = d \} \end{aligned}$$

- $A = \left[\begin{array}{ccc} a_1 & \cdots & a_m \end{array} \right]^T \in \mathbf{R}^{m \times n}$
- $C = \left[\begin{array}{ccc} c_1 & \cdots & c_p \end{array} \right]^T \in \mathbf{R}^{p \times n}$



example: nonnegative orthant $\mathbf{R}^n_+ = \{x \in \mathbf{R}^n \mid x \succeq 0\}$ is a polyhedron (where positive orthant $\mathbf{R}^n_{++} = \{x \in \mathbf{R}^n \mid x \succ 0\}$ is its interior)

simplex: convex hull of $k+1$ affinely independent points, *i.e.*,

$$C = \mathbf{conv}\{x_0, x_1, \ldots, x_k\}$$

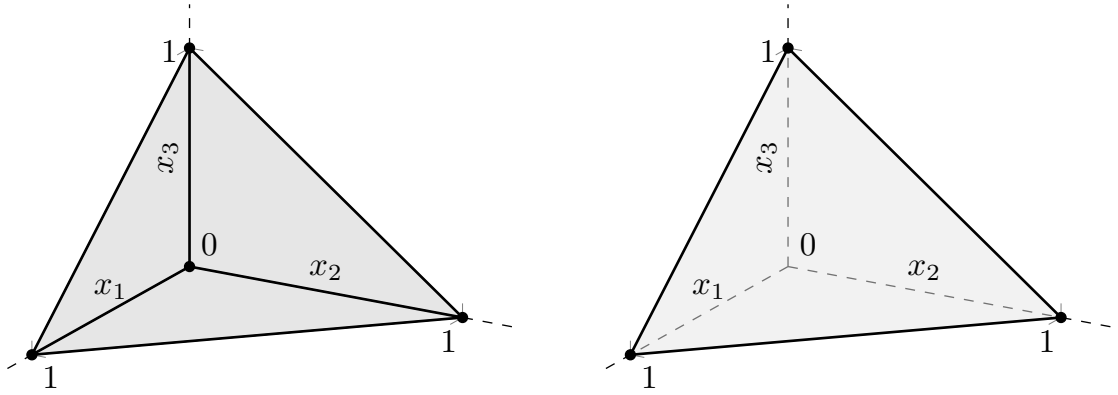
examples:

- *unit simplex* in $\mathbf{R}^n$:

$$\mathbf{conv}\{0, e_1, e_2, \ldots, e_n\} = \{x \in \mathbf{R}^n \mid x \succeq 0,\ \mathbf{1}^T x \leq 1\}$$

- *probability simplex* in $\mathbf{R}^n$:

$$\mathbf{conv}\{e_1, e_2, \ldots, e_n\} = \{x \in \mathbf{R}^n \mid x \succeq 0,\ \mathbf{1}^T x = 1\}$$



Norm balls

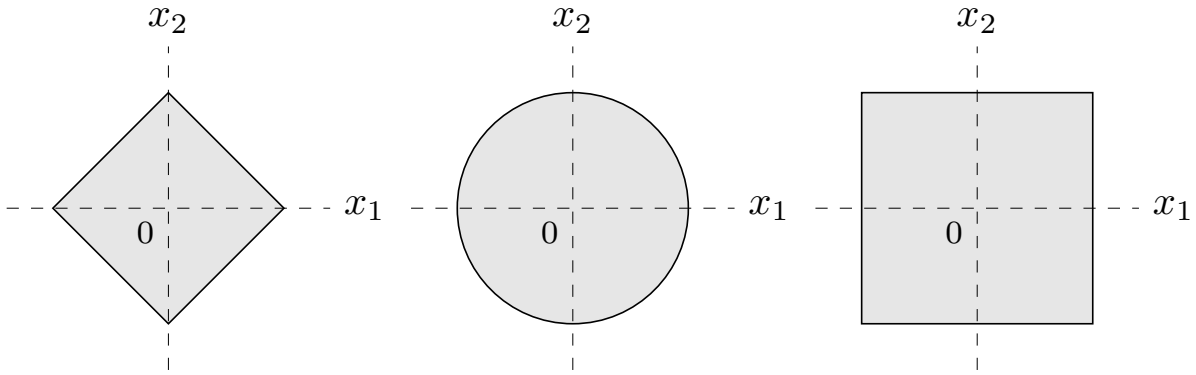
norm: a function $\|\cdot\|: \mathbf{R}^n \to \mathbf{R}$ satisfying

- $\|x\| \geq 0$ for all $x \in \mathbf{R}^n$, and $\|x\| = 0$ if and only if $x = 0$
- $\|tx\| = |t|\|x\|$ for all $x \in \mathbf{R}^n$ and $t \in \mathbf{R}$
- $\|x + y\| \leq \|x\| + \|y\|$ for all $x, y \in \mathbf{R}^n$

we use $\|\cdot\|$ for general norm, and $\|\cdot\|_{\mathrm{symb}}$ for specific norm

norm ball of radius r and center x_c: $\{x \in \mathbf{R}^n \mid \|x - x_c\| \leq r\}$

examples: unit balls of ℓ_1-, ℓ_2- (Euclidean), and ℓ_∞-norms in $\mathbf{R}^2$:



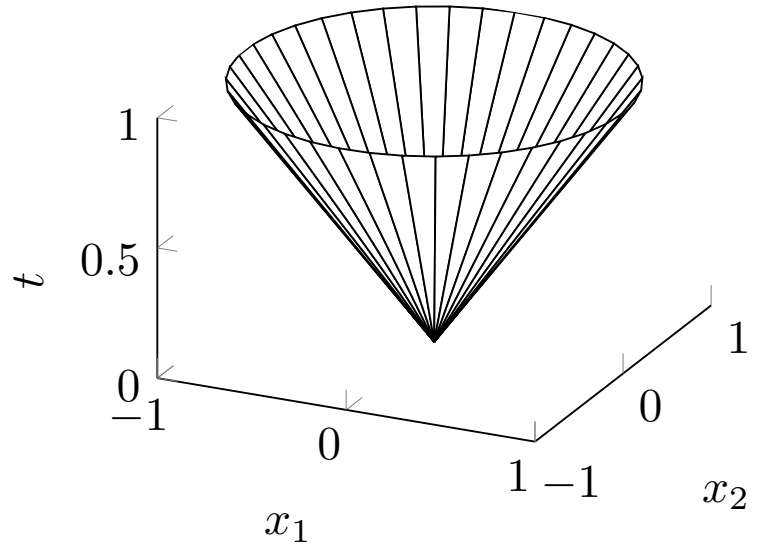
Norm cones

norm cone associated with norm $\|\cdot\|$:

$$\{(x,t) \in \mathbf{R}^n \times \mathbf{R} \mid \|x\| \leq t\}$$

- norm cones are convex cones
- example: **second-order cone** in $\mathbf{R}^3$:

$$\{(x,t) \in \mathbf{R}^2 \times \mathbf{R} \mid \|x\|_2 \leq t\}$$



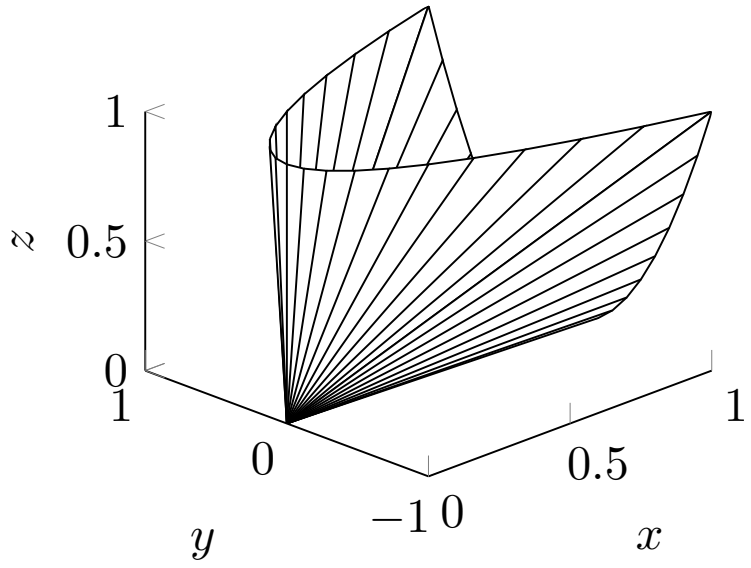
Positive semidefinite cones

- $\mathbf{S}^n$: set of symmetric $n \times n$ matrices
- $\mathbf{S}^n_+ = \{X \in \mathbf{S}^n \mid X \succeq 0\}$: positive semidefinite $n \times n$ matrices, *i.e.*,

$$X \in \mathbf{S}^n_+ \iff z^T X z \geq 0 \text{ for all } z \in \mathbf{R}^n$$

  - $\mathbf{S}^n_+$ is a convex cone
- $\mathbf{S}^n_{++} = \{X \in \mathbf{S}^n \mid X \succ 0\}$: positive definite $n \times n$ matrices

example: $\begin{bmatrix} x & y \\ y & z \end{bmatrix} \in \mathbf{S}^2_+$



Algebra of convex sets

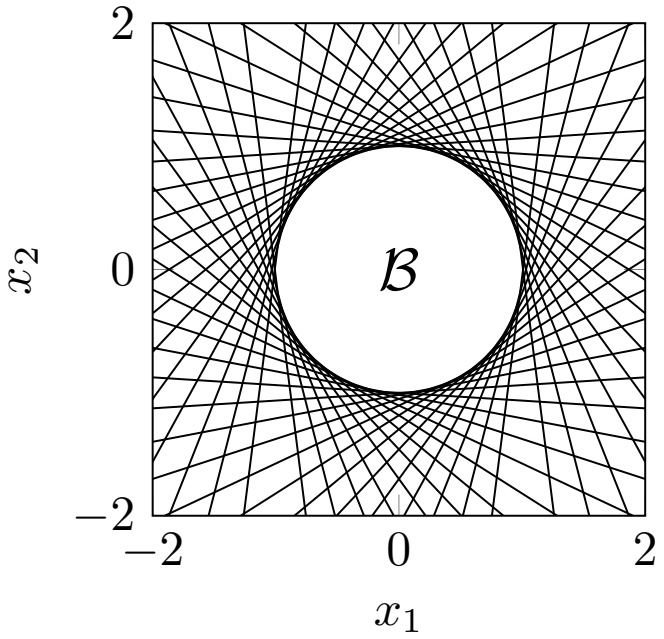
intersection of convex sets is convex, $i.e.$,

- $C_1, C_2, \ldots, C_k$ convex $\implies \bigcap_{i=1}^k C_i$ convex
- $\{C_i\}_{\mathcal{I}}$ set of convex sets indexed by $\mathcal{I}$ $\implies \bigcap_{i\in\mathcal{I}} C_i$ convex

example: unit Euclidean ball in $\mathbf{R}^2$

$$\mathcal{B} = \bigcap_{t\in[0,2\pi)} \{x \mid x_1 \cos t + x_2 \sin t \leq 1\}$$

is intersection of infinite halfspaces



Cartesian product of convex sets is convex, *i.e.*,

$$C \times D = \{(x, y) \mid x \in C,\ y \in D\}$$

is convex if C and D are convex

affine transformation: suppose $f : \mathbf{R}^n \to \mathbf{R}^m$ is affine, *i.e.*,

$$f(x) = Ax + b \text{ with } A \in \mathbf{R}^{m \times n},\ b \in \mathbf{R}^m$$

- the image of a convex set under f is convex, *i.e.*,

$$C \subseteq \mathbf{R}^n \text{ convex} \implies f(C) = \{f(x) \mid x \in C\} \text{ convex}$$

- the inverse image $f^{-1}(C)$ of a convex set under f is convex, *i.e.*,

$$C \subseteq \mathbf{R}^m \text{ convex} \implies f^{-1}(C) = \{x \in \mathbf{R}^n \mid f(x) \in C\} \text{ convex}$$

examples:

- scaling and translation of a convex set is convex, $i.e.$,

$$C \subseteq \mathbf{R}^n \text{ convex}, \quad \lambda \in \mathbf{R}, \quad v \in \mathbf{R}^n$$
$$\Longrightarrow \quad \lambda C + v = \{\lambda x + v \mid x \in C\} \text{ convex}$$

- (Minkowski) addition of convex sets is convex, $i.e.$,

$$C, D \subseteq \mathbf{R}^n \text{ convex}$$
$$\Longrightarrow \quad C + D = \{x + y \mid x \in C,\ y \in D\} \text{ convex}$$

- solution set of linear matrix inequality

$$\{x \in \mathbf{R}^n \mid x_1 A_1 + \cdots + x_n A_n \preceq B\}$$

  with $A_1, \ldots, A_n, B \in \mathbf{S}^m$ is convex

- an ellipsoid

$$\begin{aligned}\mathcal{E} &= \{x \in \mathbf{R}^n \mid (x - x_0)^T P^{-1}(x - x_0) \leq 1\} \\ &= \{x \in \mathbf{R}^n \mid \|P^{-1/2}(x - x_0)\|_2 \leq 1\}\end{aligned}$$

with center $x_0 \in \mathbf{R}^n$ and $P \in \mathbf{S}^n_{++}$ is the inverse image of the unit ball

$$\{x \in \mathbf{R}^n \mid \|x\|_2 \leq 1\}$$

under the affine transformation

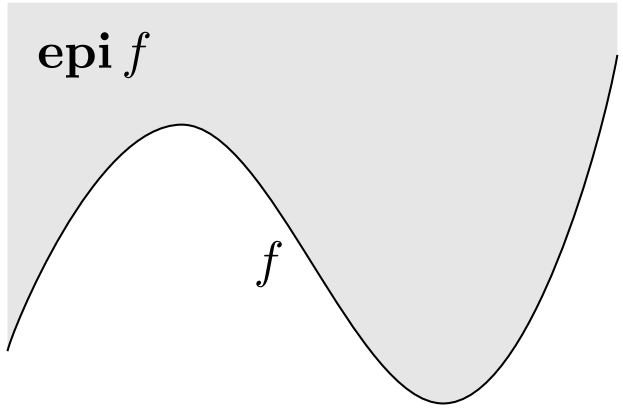
$$f(x) = P^{-1/2}(x - x_0)$$

and hence is convex

Convex functions

epigraph of function $f : \mathbf{R}^n \to \mathbf{R}$:

$$\mathbf{epi}\, f = \left\{ (x,t) \in \mathbf{R}^{n+1} \;\middle|\; \begin{array}{l} f(x) \le t \\ x \in \mathbf{dom}\, f \end{array} \right\}$$



- $f : \mathbf{R}^n \to \mathbf{R}$ is **convex** if $\mathbf{epi}\, f$ is convex
- $f : \mathbf{R}^n \to \mathbf{R}$ is **concave** if $-f$ is convex, *i.e.*, its *hypograph*

$$\mathbf{hypo}\, f = \{(x,t) \in \mathbf{R}^{n+1} \mid f(x) \ge t,\ x \in \mathbf{dom}\, f\}$$

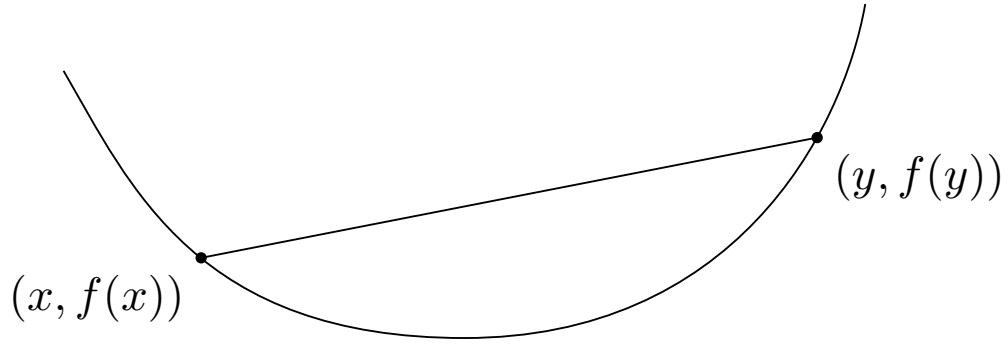
is convex

Jensen's inequality

$f: \mathbf{R}^n \to \mathbf{R}$ is convex if $\mathbf{dom}\, f$ is a convex set and

$$f(\theta x + (1-\theta)y) \leq \theta f(x) + (1-\theta) f(y)$$

for all $x, y \in \mathbf{dom}\, f$ and $\theta \in [0,1]$.



- f is strictly convex if $\mathbf{dom}\, f$ is convex and

$$f(\theta x + (1-\theta)y) < \theta f(x) + (1-\theta) f(y)$$

for all $x, y \in \mathbf{dom}\, f$, $x \neq y$, and $\theta \in (0,1)$.

extensions: let $f : \mathbf{R}^n \to \mathbf{R}$ be convex

- $x_1, \ldots, x_k \in \mathbf{dom}\, f$, $\theta \in \mathbf{R}^k$ with $\mathbf{1}^T\theta = 1$, $\theta \succeq 0 \implies$

$$f(\theta_1 x_1 + \cdots + \theta_k x_k) \leq \theta_1 f(x_1) + \cdots + \theta_k f(x_k)$$

- $p : S \to \mathbf{R}$, $S \subseteq \mathbf{dom}\, f$, $\int_S p(x)\, dx = 1$, $p(x) \geq 0$ for all $x \in S \implies$

$$f\left(\int_S p(x)x\, dx\right) \leq \int_S p(x)f(x)\, dx$$

- random vector $z \in \mathbf{R}^n$ with $\mathbf{prob}(z \in \mathbf{dom}\, f) = 1 \implies$

$$f(\mathbf{E}\, z) \leq \mathbf{E}\, f(z)$$

basic inequality is a special case with discrete distribution

$$\mathbf{prob}(z = x) = \theta, \quad \mathbf{prob}(z = y) = 1 - \theta$$

Sublevel sets

α-**sublevel set** of function $f : \mathbf{R}^n \to \mathbf{R}$:

$$C_\alpha = \{x \in \mathbf{dom}\, f \mid f(x) \leq \alpha\}$$

- f is convex $\implies$ C_α is convex for all $\alpha \in \mathbf{R}$

- converse is false: C_α convex for all $\alpha \in \mathbf{R}$ does not imply f convex
  - $f(x) = x^3$ has convex sublevel sets $(-\infty, \alpha^{1/3})$ but is not convex
  - $f(x) = \log x$ has convex sublevel sets $(0, e^\alpha]$ but is not convex

Convex envelope

suppose $F \subseteq \mathbf{R}^{n+1}$ is convex, the function $f\colon \mathbf{R}^n \to \mathbf{R}$ defined by

$$f(x) = \inf\{t \mid (x,t) \in F\}$$

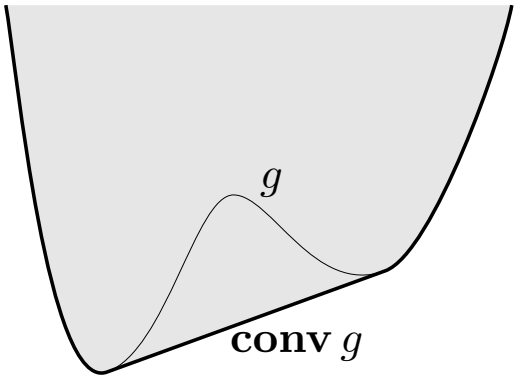
is a convex function with $\mathbf{epi}\, f = F$

convex envelope of $g\colon \mathbf{R}^n \to \mathbf{R}$: the function $f\colon \mathbf{R}^n \to \mathbf{R}$ obtained from the operation above with

$$F = \mathbf{conv}\, \mathbf{epi}\, g$$

- denoted $\mathbf{conv}\, g$
- largest convex global underestimator of g, *i.e.*,

$$\mathbf{conv}\, g(x) = \sup \left\{ h(x) \;\middle|\; \begin{array}{l} h \text{ convex} \\ h(z) \leq g(z) \\ \text{for all } z \in \mathbf{dom}\, g \end{array} \right\}$$

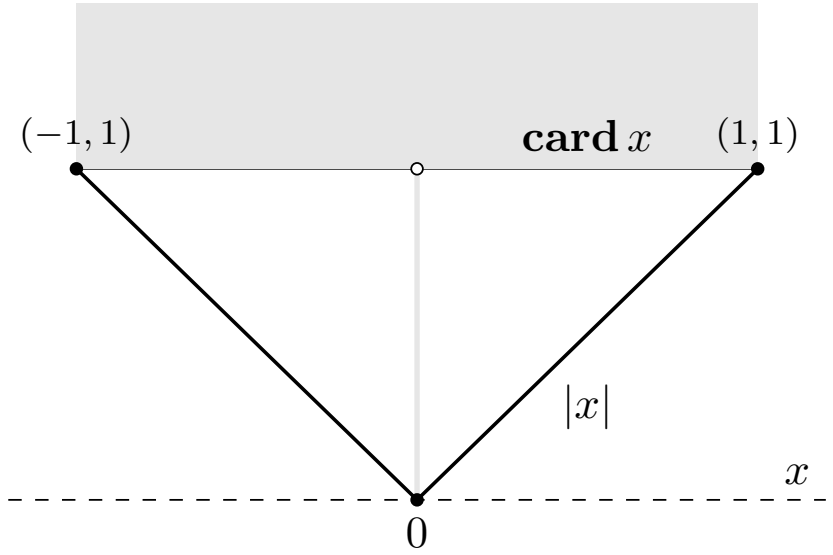


example: convex envelope of the *cardinality function* for $x \in \mathbf{R}$:

$$\mathbf{card}\, x = \begin{cases} 0, & x = 0 \\ 1, & x \neq 0 \end{cases}$$

on the interval $[-1, 1]$ is the absolute value function, *i.e.*,

$$\mathbf{conv}\, \mathbf{card}\, x = |x|$$

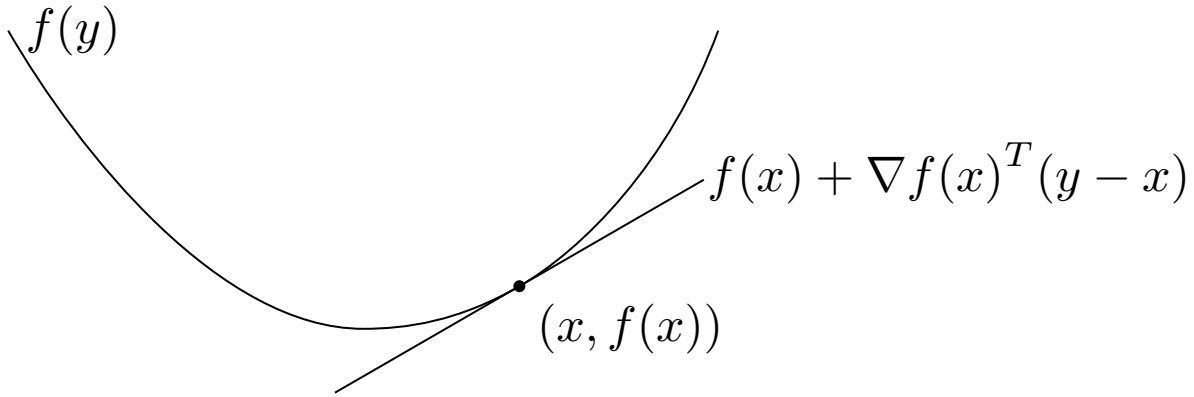


First-order conditions

differentiable $f : \mathbf{R}^n \to \mathbf{R}$ with convex domain is convex iff

$$f(y) \geq f(x) + \nabla f(x)^T (y - x) \quad \text{for all } x, y \in \mathbf{dom}\, f$$

where $\nabla f(x) = \left(\frac{\partial f}{\partial x_1}, \ldots, \frac{\partial f}{\partial x_n} \right)$ is the gradient of f at $x \in \mathbf{R}^n$



- 1st-order approximation of convex f is global underestimator
- strictly convex f if $>$ holds for $x \neq y$ (converse also true)

Second-order conditions

twice differentiable $f : \mathbf{R}^n \to \mathbf{R}$ with convex domain is convex iff

$$\nabla^2 f(x) \succeq 0 \quad \text{for all } x \in \mathbf{dom}\, f$$

where $\nabla^2 f(x) \in \mathbf{S}^n$ is the Hessian of f at $x \in \mathbf{R}^n$, *i.e.*,

$$\nabla^2 f(x)_{ij} = \frac{\partial^2 f(x)}{\partial x_i \partial x_j}, \quad i, j = 1, \ldots, n$$

- strictly convex f if $\nabla^2 f(x) \succ 0$ for all $x \in \mathbf{dom}\, f$
- converse not true: $f(x) = x^4$ is strictly convex but $\nabla^2 f(0) = 0$

convex domain cannot be dropped from 1st- and 2nd-order conditions

e.g., $f(x) = 1/x^2$ nonconvex but $\nabla^2 f(x) \succ 0$ for all $x \neq 0$

Examples on $\mathbf{R}$

convex:

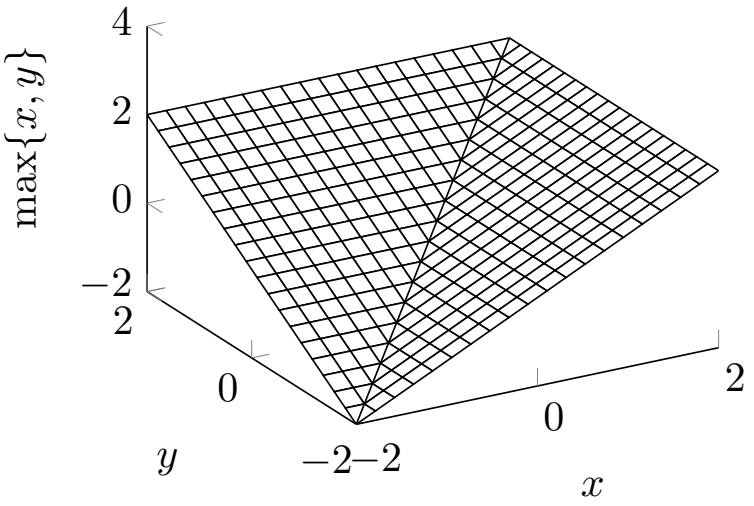
- affine: $f(x) = ax + b$ on $\mathbf{R}$ for any $a, b \in \mathbf{R}$
- quadratic: $f(x) = ax^2 + bx + c$ on $\mathbf{R}$ for any $a, b, c \in \mathbf{R}$, $a > 0$
- absolute value: $f(x) = |x|$ on $\mathbf{R}$
- powers: $f(x) = x^p$ on $\mathbf{R}_{++}$ for $p \leq 0$ or $p \geq 1$
- exponential: $f(x) = e^x$ on $\mathbf{R}$
- negative entropy: $f(x) = x \log x$ on $\mathbf{R}_{++}$

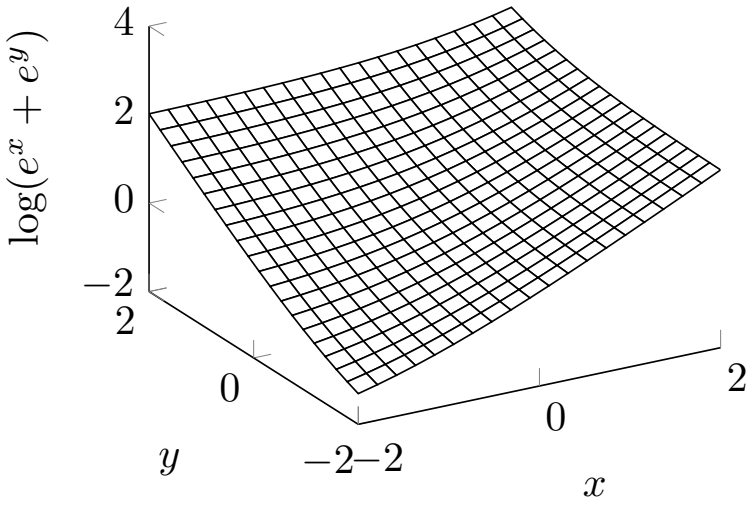
concave:

- affine: $f(x) = ax + b$ on $\mathbf{R}$ for any $a, b \in \mathbf{R}$
- powers: $f(x) = x^p$ on $\mathbf{R}_{++}$ for $0 < p \leq 1$
- logarithm: $f(x) = \log x$ on $\mathbf{R}_{++}$

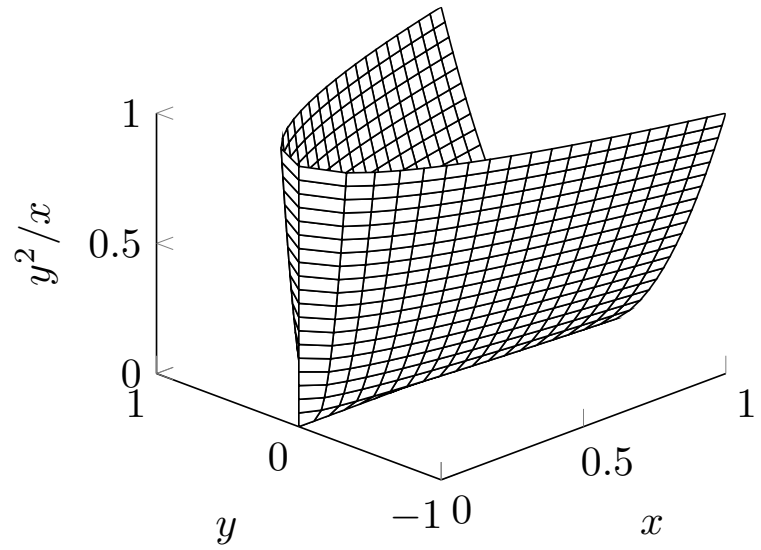
Examples on $\mathbf{R}^n$

- affine: $f(x) = a^T x + b$ convex and concave on $\mathbf{R}^n$
- quadratic: $f(x) = x^T P x + q^T x + r$ convex on $\mathbf{R}^n$ for $P \succeq 0$
- norms: $f(x) = \|x\|$ convex on $\mathbf{R}^n$ for any norm $\|\cdot\|$
  e.g., $\|x\|_p = (\sum_{i=1}^n |x_i|^p)^{1/p}$ for $p \geq 1$; $\|x\|_\infty = \max_{i=1,\ldots,n} |x_i|$
- max: $f(x) = \max\{x_1, \ldots, x_n\}$ convex on $\mathbf{R}^n$
- log-sum-exp: $f(x) = \log \sum_{i=1}^n e^{x_i}$ convex on $\mathbf{R}^n$





- quadratic-over-linear: $f(x,y) = y^2/x$ convex on $\mathbf{R}_{++} \times \mathbf{R}$



- relative entropy: $f(x,y) = \sum_{i=1}^n x_i \log(x_i/y_i)$ convex on $\mathbf{R}^n_{++} \times \mathbf{R}^n_{++}$
- Kullback-Leibler (KL) divergence:

$$f(x,y) = \sum_{i=1}^n (x_i \log(x_i/y_i) - x_i + y_i)$$

convex on $\mathbf{R}^n_{++} \times \mathbf{R}^n_{++}$

Examples on $\mathbf{R}^{m\times n}$

- affine:

$$f(X) = \mathbf{tr}(A^T X) + b = \sum_{i=1}^{m}\sum_{j=1}^{n} A_{ij}X_{ij} + b$$

  convex and concave on $\mathbf{R}^{m\times n}$ for any $A \in \mathbf{R}^{m\times n}$, $b \in \mathbf{R}$

- spectral norm:

$$f(X) = \|X\|_2 = \sigma_{\max}(X) = (\lambda_{\max}(X^T X))^{1/2}$$

  convex on $\mathbf{R}^{m\times n}$ ($\sigma_{\max}$, $\lambda_{\max}$: largest singular value, eigenvalue)

- log-determinant:

$$f(X) = \log\det X$$

  concave on $\mathbf{S}^n_{++}$ (analogous to $f(x) = \log x$ on $\mathbf{R}_{++}$)

Probability distributions

logarithm of many probability distributions are concave ($i.e.$, log-concave):

- multivariate normal:

$$f(x) = \frac{1}{\sqrt{(2\pi)^n \det \Sigma}} \exp\left(-\frac{1}{2}(x-\mu)^T \Sigma^{-1}(x-\mu)\right)$$

  where $\mu \in \mathbf{R}^n$ is the mean and $\Sigma \in \mathbf{S}^n_{++}$ is the covariance matrix

- uniform:

$$f(x) = \begin{cases} 1/\alpha, & x \in C \\ 0, & \text{otherwise} \end{cases}$$

  where $C \subseteq \mathbf{R}^n$ is a convex set and $\alpha > 0$ is its volume; log given by

$$\log f(x) = \begin{cases} -\log \alpha, & x \in C \\ -\infty, & \text{otherwise} \end{cases}$$

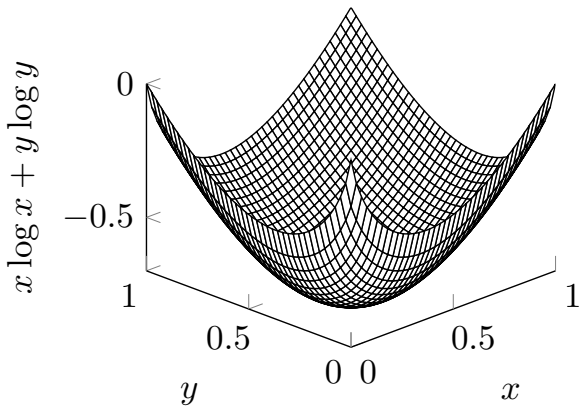
Basic functional operations

nonnegative weighted sum of convex functions is convex, $i.e.$,

$$\begin{array}{l} f_1, \ldots, f_k \text{ convex} \\ w_1, \ldots, w_k \geq 0 \end{array} \quad \Longrightarrow \quad w_1 f_1 + \cdots + w_k f_k \text{ convex}$$

- example: negative entropy

$$f(x) = \sum_{i=1}^{n} x_i \log x_i$$



  convex on $\mathbf{R}^n_{++}$

- extension: $f(x, y)$ convex in x, $w(y) \geq 0$ for all fixed $y \in S$ $\Longrightarrow$

$$g(x) = \int_S w(y) f(x, y)\, dy$$

  convex in x (basic property as discrete case)

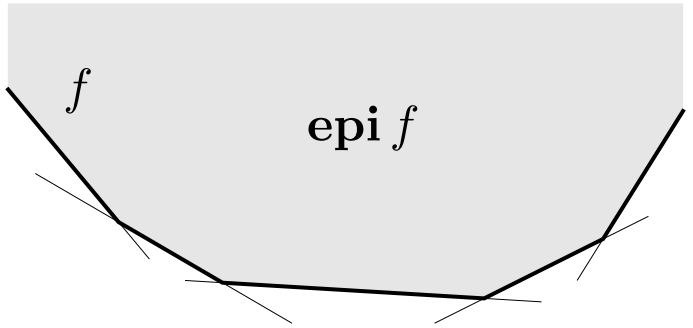
pointwise maximum of convex functions is convex, $i.e.$,

$$f_1, \ldots, f_k \text{ convex} \quad \Longrightarrow \quad f(x) = \max\{f_1(x), \ldots, f_k(x)\} \text{ convex}$$

- $\mathbf{dom}\, f = \bigcap_{i=1}^k \mathbf{dom}\, f_i$
- proof via epigraph: $\mathbf{epi}\, f = \bigcap_{i=1}^k \mathbf{epi}\, f_i$ is intersection of convex sets
- example: piecewise linear function

$$f(x) = \max_{i=1,\ldots,k} (a_i^T x + b_i) \quad (a_i \neq 0)$$

convex on $\mathbf{R}^n$



- pointwise minimum of concave functions is concave

pointwise supremum: $\{f_i\}_{i\in\mathcal{I}}$ convex $\Longrightarrow$

$$f(x) = \sup_{i\in\mathcal{I}} f_i(x)$$

with

$$\mathbf{dom}\, f = \left\{ x \in \bigcap_{i\in\mathcal{I}} \mathbf{dom}\, f_i \;\middle|\; \sup_{i\in\mathcal{I}} f_i(x) < \infty \right\}$$

is convex

- example: distance to the farthest point in a set $C \subseteq \mathbf{R}^n$:

  $$f(x) = \sup_{y\in C} \|x - y\|$$

  is convex on $\mathbf{R}^n$
- corollary: any convex function $f : \mathbf{R}^n \to \mathbf{R}$ can be expressed as

  $$f(x) = \sup\{a^T x + b \mid a^T y + b \le f(y) \text{ for all } y \in \mathbf{R}^n\}$$

  $i.e.$, the supremum of all its affine underestimators

Composition with affine

given $f: \mathbf{R}^m \to \mathbf{R}$, then $g: \mathbf{R}^n \to \mathbf{R}$ with

$$g(x) = f(Ax + b), \quad \mathbf{dom}\, g = \{x \mid Ax + b \in \mathbf{dom}\, f\}$$

has the same convexity as f for any $A \in \mathbf{R}^{m \times n}$, $b \in \mathbf{R}^m$

examples:

- log-barrier function for linear inequalities $a_i^T x \leq b_i$, $i = 1, \ldots, m$:

$$f(x) = -\sum_{i=1}^{m} \log(b_i - a_i^T x)$$

  convex on $\mathbf{dom}\, f = \{x \mid a_i^T x < b_i,\ i = 1, \ldots, m\}$
- linear approximation cost $f(x) = \|Ax - b\|$ convex on $\mathbf{R}^n$ for any norm
- least squares cost $f(x) = \|Ax - b\|_2^2$ convex on $\mathbf{R}^n$

General composition

given $h\colon \mathbf{R}^k \to \mathbf{R}$ and $g_i\colon \mathbf{R}^n \to \mathbf{R}$, $i = 1, \ldots, k$, define $f\colon \mathbf{R}^n \to \mathbf{R}$ as

$$f(x) = h(g_1(x), \ldots, g_k(x))$$

where

$$\mathbf{dom}\, f = \left\{ x \in \mathbf{R}^n \;\middle|\; \begin{array}{l} x \in \mathbf{dom}\, g_i,\ i = 1, \ldots, k \\ (g_1(x), \ldots, g_k(x)) \in \mathbf{dom}\, h \end{array} \right\}$$

- f convex if $\mathbf{dom}\, f$ convex, h convex, and for $i = 1, \ldots, k$:
  - g_i affine, or
  - h nondecreasing in its ith argument, and g_i convex, or
  - h nonincreasing in its ith argument, and g_i concave

- similar conditions holds for concavity of f

proof for $k = n = 1$: let $f = h \circ g$ with $\mathbf{dom}\, f$ convex,

$$f''(x) = h''(g(x))g'(x)^2 + h'(g(x))g''(x)$$

- assume h convex and nondecreasing, g convex
- $\Longrightarrow$ $h''(g(x)) \geq 0$, $h'(g(x)) \geq 0$, $g''(x) \geq 0$ for all $x \in \mathbf{dom}\, f$
- $\Longrightarrow$ $f''(x) \geq 0$ for all $x \in \mathbf{dom}\, f$, *i.e.*, f convex

examples:

- $f(x) = |x|^p$ convex on $\mathbf{R}$ for $p \geq 1$
- $f(x) = 1/g(x)$ convex if g concave and positive
- $f(x) = \exp g(x)$ convex if g convex
- $f(x) = \sum_{i=1}^k \log g_i(x)$ concave if g_i concave and positive
- $f(x) = \left(\sum_{i=1}^k g_i(x)^p\right)^{1/p}$ $(p \geq 1)$ convex if g_i convex and nonnegative
- $f(x) = \log \sum_{i=1}^k \exp g_i(x)$ convex if g_i convex

Optimization problem in standard form

$$\begin{array}{ll} \text{minimize} & f_0(x) \\ \text{subject to} & f_i(x) \leq 0, \quad i = 1, \ldots, m \\ & h_i(x) = 0, \quad i = 1, \ldots, p \end{array}$$

- $x \in \mathbf{R}^n$: optimization or decision *variable*
- $f_0 : \mathbf{R}^n \to \mathbf{R}$: *objective* function
- $f_1, \ldots, f_m : \mathbf{R}^n \to \mathbf{R}$: *inequality constraint* functions
- $h_1, \ldots, h_p : \mathbf{R}^n \to \mathbf{R}$: *equality constraint* functions

goal: find x that minimizes $f_0(x)$ while satisfying the constraints

Optimal value and points

optimal value of a problem:

$$p^\star = \inf\{f_0(x) \mid x \in \mathcal{D}\}$$

where $\mathcal{D}$ is the **feasible set**:

$$\mathcal{D} = \left\{ x \in \bigcap_{i=0}^{m} \mathbf{dom}\, f_i \cap \bigcap_{i=1}^{p} \mathbf{dom}\, h_i \;\middle|\; \begin{array}{l} f_i(x) \leq 0,\ i = 1, \ldots, m \\ h_i(x) = 0,\ i = 1, \ldots, p \end{array} \right\}$$

- $p^\star = \infty$: problem is **infeasible**, *i.e.*, $\mathcal{D} = \emptyset$
- $p^\star = -\infty$: problem is **unbounded below**
- a point x is **feasible** if $x \in \mathcal{D}$, otherwise **infeasible**

a feasible point $x^\star$ is **optimal** if $f_0(x^\star) = p^\star$

x is **locally optimal** if there is $\epsilon > 0$ such that

$$f_0(x) = \inf\{f_0(z) \mid z \in \mathcal{D},\ \|z - x\|_2 \leq \epsilon\}$$

examples ($n = 1$, $m = p = 0$):

- $f_0(x) = x \log x$, $\mathbf{dom}\, f_0 = \mathbf{R}_{++}$: $p^\star = -1/e$, $x^\star = 1/e$
- $f_0(x) = 1/x$, $\mathbf{dom}\, f_0 = \mathbf{R}_{++}$: $p^\star = 0$, no optimal point
- $f_0(x) = -\log x$, $\mathbf{dom}\, f_0 = \mathbf{R}_{++}$: $p^\star = -\infty$
- $f_0(x) = x^3 - 3x$: $p^\star = -\infty$, local optimum at $x = 1$

Feasibility problem

$$\begin{array}{ll} \text{find} & x \\ \text{subject to} & f_i(x) \geq 0, \quad i = 1, \ldots, m \\ & h_i(x) = 0, \quad i = 1, \ldots, p \end{array}$$

can be considered a special case of the standard form with $f_0(x) = 0$:

$$\begin{array}{ll} \text{minimize} & 0 \\ \text{subject to} & f_i(x) \geq 0, \quad i = 1, \ldots, m \\ & h_i(x) = 0, \quad i = 1, \ldots, p \end{array}$$

- $p^\star = 0$ if constraints are feasible
- any feasible x is optimal
- $p^\star = \infty$ if constraints are infeasible

Equivalent problems

two problems are (informally) **equivalent** if the solution of one is readily obtained from the solution of the other, and vice versa

examples:

- change of variables; scaling and translation of objective
- introduce slack variables:

$$\begin{array}{ll} \text{minimize (over } x) & f_0(x) \\ \text{subject to} & f_i(x) \leq 0, \quad i = 1, \ldots, m \end{array}$$

is equivalent to

$$\begin{array}{ll} \text{minimize (over } x, s) & f_0(x) \\ \text{subject to} & f_i(x) + s_i = 0, \quad i = 1, \ldots, m \\ & s \succeq 0 \end{array}$$

where s_i is the *slack variable* associated with $f_i(x) \leq 0$

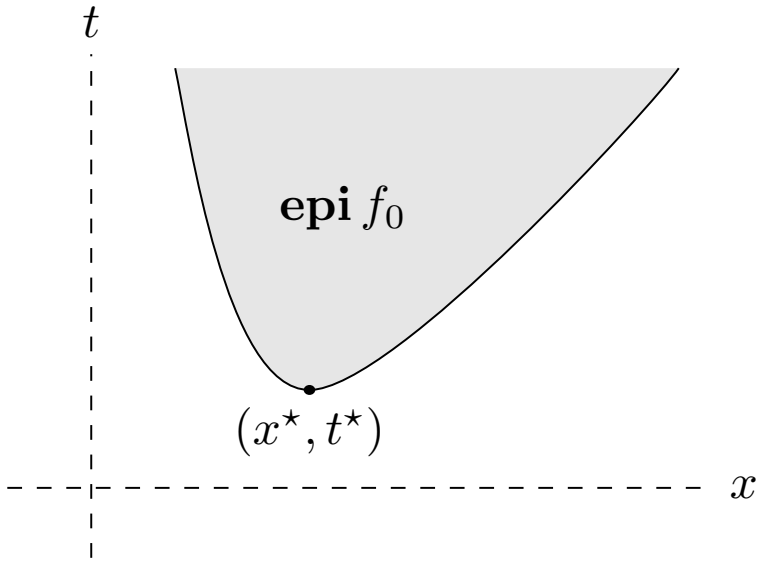
- introduce epigraph variable:

$$\text{minimize (over } x) \quad f_0(x)$$

is equivalent to

$$\begin{array}{ll} \text{minimize (over } x, t) & t \\ \text{subject to} & f_0(x) - t \leq 0 \end{array}$$

where t is the epigraph variable associated with $f_0(x)$



Relaxations

one problem is a **relaxation** of another if the feasible set of the first contains that of the second

basic properties:

- relaxation lower bounds optimal value of the original problem
- relaxation optimal point is original feasible
  $\Longrightarrow$ it is optimal for the original problem

example: the combinatorial problem

$$\begin{array}{ll} \text{minimize} & f_0(x_1, \ldots, x_m) \\ \text{subject to} & x_i \in \{0,1\}^n, \quad \mathbf{card}\, x_i = 1, \quad i = 1, \ldots, m \end{array}$$

is often relaxed to

$$\begin{array}{ll} \text{minimize} & f_0(x_1, \ldots, x_m) \\ \text{subject to} & x_i \succeq 0, \quad \mathbf{1}^T x_i = 1, \quad i = 1, \ldots, m \end{array}$$

Convex optimization problem

standard form of a convex optimization problem:

$$\begin{array}{ll} \text{minimize} & f_0(x) \\ \text{subject to} & f_i(x) \leq 0, \quad i = 1, \ldots, m \\ & a_i^T x - b_i = 0, \quad i = 1, \ldots, p \end{array}$$

- $f_0, f_1, \ldots, f_m$ convex functions
- equality constraints are affine
- often written as

$$\begin{array}{ll} \text{minimize} & f_0(x) \\ \text{subject to} & f_i(x) \leq 0, \quad i = 1, \ldots, m \\ & Ax = b \end{array}$$

- important property: feasible set of a convex problem is convex

example:

$$\begin{array}{ll} \text{minimize} & f_0(x) = x_1^2 + x_2^2 \\ \text{subject to} & f_1(x) = x_1/(1+x_2^2) \leq 0 \\ & h_1(x) = (x_1+x_2)^2 = 0 \end{array}$$

- f_0 convex; feasible set $\{(x_1, x_2) \mid x_1 = x_2 \leq 0\}$ convex

- (by definition) *not* a convex problem: f_1 not convex, h_1 not affine

- equivalent (but not identical) to convex problem:

$$\begin{array}{ll} \text{minimize} & x_1^2 + x_2^2 \\ \text{subject to} & x_1 \leq 0 \\ & x_1 + x_2 = 0 \end{array}$$

Optimal points

any locally optimal point of a convex problem is global optimal

proof: let $x \in \mathcal{D}$ be locally optimal, *i.e.*, for some $\epsilon > 0$,

$$f_0(x) = \inf\{f_0(z) \mid z \in \mathcal{D},\ \|z - x\|_2 \leq \epsilon\}$$

- suppose $y \in \mathcal{D}$ is feasible with $f_0(y) < f_0(x)$, then $\|y - x\|_2 > \epsilon$
- $\mathcal{D}$ convex $\implies$ $z = \theta x + (1-\theta)y \in \mathcal{D}$ for all $\theta \in (0,1)$
- choose $\theta = \epsilon/(2\|y - x\|_2) \in (0,1)$, then

$$\|z - x\|_2 = \epsilon/2 < \epsilon \quad \implies \quad f_0(z) \geq f_0(x)$$

- from convexity of f_0:

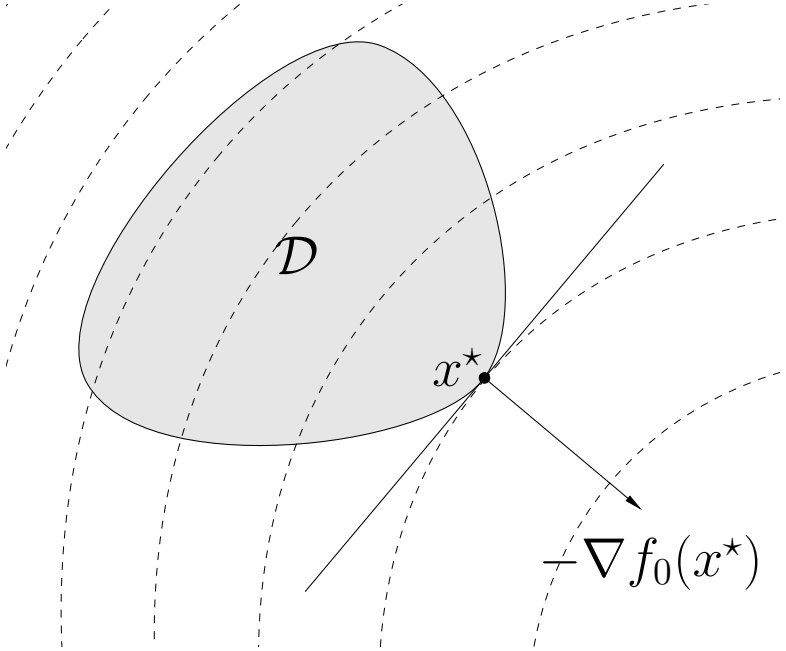
$$f_0(z) = f_0(\theta x + (1-\theta)y) \leq \theta f_0(x) + (1-\theta) f_0(y) < f_0(x)$$

contradiction $\implies$ no such y exists, *i.e.*, x is globally optimal

Optimality conditions

for differentiable f_0, $x \in \mathcal{D}$ is optimal $\iff$

$$\nabla f_0(x)^T(y - x) \geq 0 \text{ for all } y \in \mathcal{D}$$

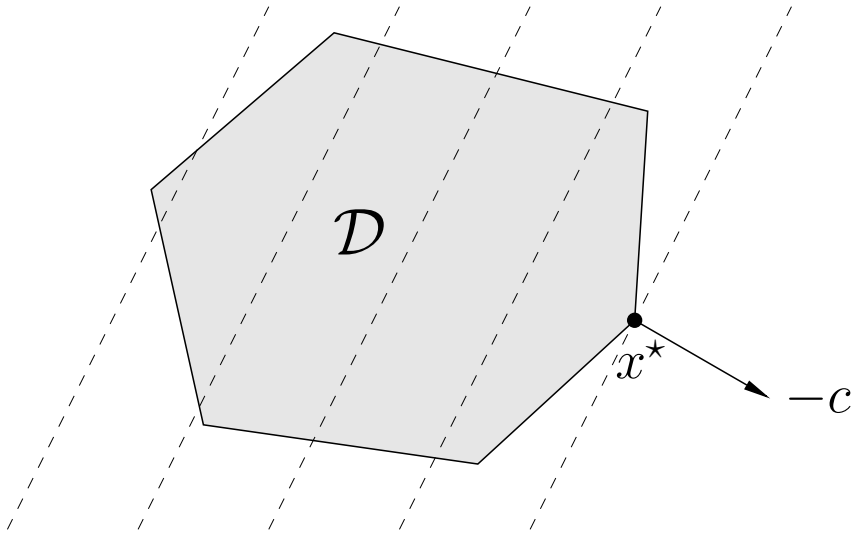


- nonzero $\nabla f_0(x)$ defines a *supporting hyperplane* to $\mathcal{D}$ at x
- unconstrained problem: x optimal $\iff$ $x \in \mathbf{dom}\, f_0$, $\nabla f_0(x) = 0$

Linear program (LP)

$$\begin{array}{ll} \text{minimize} & c^T x \\ \text{subject to} & Gx \preceq h \\ & Ax = b \end{array}$$

- convex problem with affine objective and constraint functions
- feasible set is a polyhedron



example: least ℓ_1-norm fitting

$$\begin{array}{ll} \text{minimize} & \|x\|_1 \\ \text{subject to} & Ax = b \end{array}$$

- equivalent to the LP

$$\begin{array}{ll} \text{minimize} & \mathbf{1}^T t \\ \text{subject to} & -t \preceq x \preceq t \\ & Ax = b, \end{array}$$

  with variables x and t

- another equivalent LP formulation:

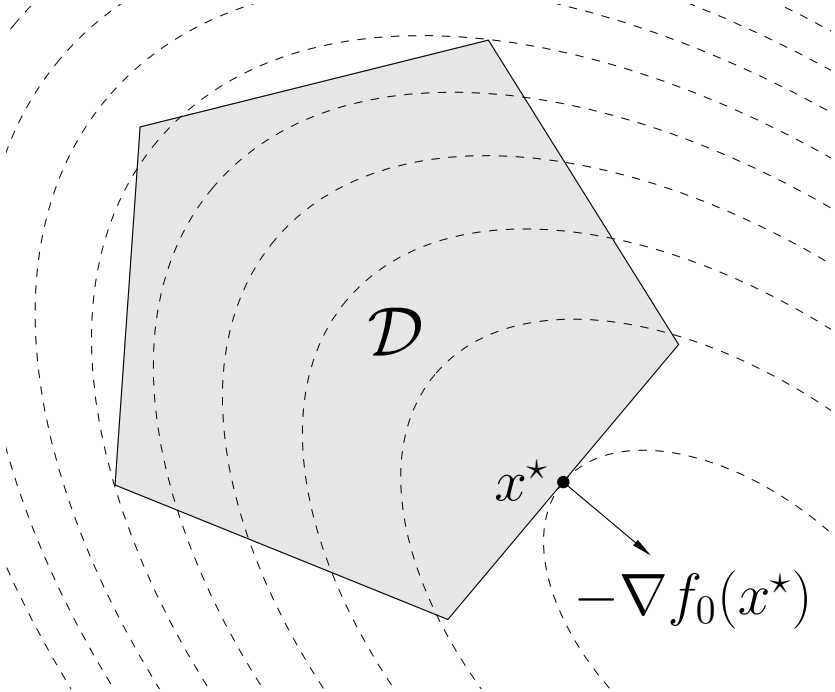
$$\begin{array}{ll} \text{minimize} & \mathbf{1}^T x_+ + \mathbf{1}^T x_- \\ \text{subject to} & x_+ \succeq 0, \quad x_- \succeq 0 \\ & A(x_+ - x_-) = b \end{array}$$

  with variables $x_+, x_- \in \mathbf{R}^n$

Quadratic program (QP)

$$\begin{array}{ll} \text{minimize} & (1/2)x^T P x + q^T x + r \\ \text{subject to} & Gx \preceq h \\ & Ax = b \end{array}$$

- $P \in \mathbf{S}_+^n$ so objective is a convex quadratic function



quadratically constrained quadratic program (QCQP):

$$\begin{array}{ll} \text{minimize} & (1/2)x^T P_0 x + q_0^T x + r_0 \\ \text{subject to} & (1/2)x^T P_i x + q_i^T x + r_i \leq 0, \quad i = 1, \ldots, m \\ & Ax = b \end{array}$$

- $P_0, \ldots, P_m \in \mathbf{S}_+^n$ so objective and constraints are convex quadratics
- if $P_1, \ldots, P_m \in \mathbf{S}_{++}^n$, then feasible set is intersection of m ellipsoids and an affine set
- QCQP reduces to
  - QP if $P_1, \ldots, P_m = 0$
  - LP if $P_0, P_1, \ldots, P_m = 0$

examples:

- least squares problem

$$\text{minimize} \quad \|Ax - b\|_2^2 = x^T A^T A x - 2b^T A x + b^T b$$

  is a QP with $P = 2A^T A$, $q = -2A^T b$, $r = b^T b$

- norm bounded least squares problem

$$\begin{array}{ll} \text{minimize} & \|Ax - b\|_2^2 \\ \text{subject to} & \|x\|_2 \leq 1 \end{array}$$

  is equivalent to the QCQP

$$\begin{array}{ll} \text{minimize} & x^T A^T A x - 2b^T A x + b^T b \\ \text{subject to} & x^T x - 1 \leq 0 \end{array}$$

Semidefinite program (SDP)

$$\begin{array}{ll} \text{minimize} & c^T x \\ \text{subject to} & x_1 F_1 + \cdots + x_n F_n + G \preceq 0 \\ & Ax = b \end{array}$$

- $F_1, \ldots, F_n, G \in \mathbf{S}^m$
- inequality constraint is called *linear matrix inequality* (LMI)
- includes problems with multiple LMI constraints, *e.g.*,

$$x_1 F_1^{(i)} + \cdots + x_n F_n^{(i)} + G^{(i)} \preceq 0, \quad i = 1, \ldots, k$$

  is equivalent to a single LMI constraint

$$\sum_{i=1}^{n} x_i \begin{bmatrix} F_i^{(1)} & & \\ & \ddots & \\ & & F_i^{(k)} \end{bmatrix} + \begin{bmatrix} G^{(1)} & & \\ & \ddots & \\ & & G^{(k)} \end{bmatrix} \preceq 0$$

examples:

- the inequality form LP

$$\begin{array}{ll} \text{minimize} & c^T x \\ \text{subject to} & Ax \preceq b \end{array}$$

  is equivalent to the SDP

$$\begin{array}{ll} \text{minimize} & c^T x \\ \text{subject to} & \mathbf{diag}(Ax - b) \preceq 0 \end{array}$$

  (note different parsing of the inequality $\preceq$)

- optimal value of the SDP

$$\begin{array}{ll} \text{minimize} & t \\ \text{subject to} & A - tI \succeq 0 \end{array}$$

  finds the smallest eigenvalue $\lambda_{\min}(A)$ for $A \in \mathbf{S}^n$